# AlgoFlow

*One page project summary*

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**Application repository** https://github.com/KRISH141006/AlgoFlow

**Extension repository** https://github.com/KRISH141006/dsa_universal_sync

### Problem statement

Solving a DSA problem once does not ensure that its approach will be remembered. Accepted code, topic notes, and revision dates are often scattered across coding platforms and local files. AlgoFlow brings these records into one local workspace and schedules deliberate revision.

### Objectives

Organize accepted LeetCode and GeeksforGeeks solutions; preserve exact code and problem descriptions; schedule problem and theory reviews; record improvements as solution versions; and support safe backup and restoration.

### Key features

AlgoFlow uses my custom DSA Universal Sync extension to collect code and problem details from LeetCode and GeeksforGeeks. The extension can fetch solved problems together as a batch and automatically save code when a new submission is accepted. Its JSON exports can be imported into AlgoFlow. Other features include a searchable library, duplicate protection, saved solution versions, scheduled reviews, progress tracking, and backup and restore.

### Technologies and tools

Python, FastAPI, SQLite, React, TypeScript, Vite, JavaScript, browser extension APIs, Pydantic, Bleach, pytest, and GitHub.

### Implementation

The extension collects solutions and exports them as JSON. The React interface sends imported data and form entries to the FastAPI backend. The backend validates the data and stores problems, topics, solution versions, and review history in SQLite. Reviews follow intervals of 1, 7, 30, and 180 days, and saving a new version keeps earlier code available.

### Achieved outcome

AlgoFlow imports JSON files generated by the custom extension and organizes the problem details and submitted code in a searchable library. The dashboard shows due reviews and a summary of progress. The application schedules newly imported problems for revision and updates the next review date when a review is completed. Users can revisit saved solutions, record improved versions, and back up their data. This provides a clear workflow for collecting solutions and revising them regularly.